MPR Paper 18

Part A

(01) Two identical spheres A and C each of mass m are projected towards a stationery identical sphere B which is midway between them on a smooth horizontal floor with velocities u and λu respectively. A and B collide and immediately afterwards B and C collides and coalesces. If after the collision the composite particle BC starts moving with velocity $u/4$ in the direction AB, then show that the coefficient of restitution between the spheres is 2λ.

(02) A particle is projected from a point O on a horizontal ground with a velocity $u = \sqrt{2ag}$ at an angle θ to the horizontal. It just passes over a vertical obstacle of height $a/2$ at a horizontal distance a from O. Show that $\tan\theta = 1$ or $\tan\theta = 3$. Here g is the gravitational acceleration.

(03) A particle of mass m is placed at a point B in a smooth horizontal groove made in a fixed wedge. A particle of mass 2m is placed at point A on a smooth plane inclined at an angle 30° to the horizontal and is connected by a light inextensible string to another particle as shown in the figure. The system is released from rest. Find the common acceleration of the particles. If the particle of mass m is given a horizontal impulse $m\sqrt{\frac{2ga}{3}}$ when it reaches C, show that the particle of mass m comes to instantaneous rest.

(04) A lorry of mass 5 metric ton has a maximum speed of $10\ ms^{-1}$ when travelling on a horizontal road. If the power of the engine is 25 kW find the road resistance. Find the maximum speed at which this lorry can travel up an incline of $\sin^{-1}\left(\frac{1}{100}\right)$ to the horizontal against the same resistance. $(g = 10\ ms^{-2})$.

(05) A particle of mass m is attached to one end of a light rod AB, the other end A which is fixed to a point on the circumference of a circular disc of radius a which is fixed horizontally. The disc is rotating about its vertical axis with a uniform angular velocity, the particle describing a horizontal circle with the rod inclined at an angle 30° to the vertical as shown in the figure. Find the tension in the light rod and show that the ~~angu~~ angular velocity is $\sqrt{\frac{9}{2\sqrt{3}a}}$.

(06) In usual notation, Let the position vectors of two points A and B corresponding to a fixed point O are $2i+\alpha j$ and $\beta i+j$. Let C be a point such that OACB is a rhombus. Show that $\beta^2-\alpha^2=3$.

(07) A uniform rod AB of weight W is in equilibrium on two smooth planes inclined at angles 30° and 60° to the horizontal. (see figure) Mark the forces acting on the rod and find the inclination of the rod to the vertical and reactions of the planes at A and B.

(08) A particle P of mass m is placed on a rough inclined plane of inclination $\tan^{-1}(3/4)$ to the horizontal and is connected by a light inextensible string to a particle Q of of mass 2m on a smooth horizontal plane at the top of the incline as shown. The coefficient of friction between P and inclined plane is $1/3$. An elastic string of natural length l and modulus of elasticity λ is fixed to a point on a horizontal plane and attached to particle Q, the extension of the string being l when the system is in equilibrium. Show that $\frac{mg}{3} \leq \lambda \leq \frac{13mg}{15}$.

(09) A and B are two independent events in a sample space S. If $P(A)=1/3$ and $P(B)=1/4$, then show that

$P[(A\cup B)\cap(A'\cup B')]=\frac{5}{12}$.

(10) The values of six observations in ascending order are 1, 5, x, y, z, 12 where $x, y, z \in \mathbb{Z}^+$. These observations has no mode and the mean and median are equal to 7. Find the values of x, y and z and also find the variance of these observations.

Part B

11 (a) A front-open lift attached to a vertical tower is moving upwards with a uniform speed u. At a certain instant, a man inside it throws a object P vertically upwards with a velocity v $(v>u)$ relative to him, under gravity. When the Object P is instantaneously at rest relative to the ground, another object Q is released by the man from rest relative to himself. The acceleration due to gravity is $g\ ms^{-2}$.
Draw the velocity-time graph for the motion of P relative to the lift.
Hence, find the maximum seperation between them and the time taken for the particle P to be instantaneously at rest relative to the ground.
Considering the velocity-time graph for the motion of particle Q relative to the lift in the same above diagram, find the seperation between Q and the lift, when P and the lift meet again.
Also, find the time taken for P and Q to meet.

(b) A warship S is moving towards north with a uniform velocity of $\sqrt{3}u\ kmh^{-1}$. At a certain instant, two enemy boats B_1 and B_2 leave a port L, which is $4\sqrt{3}$ km west of the position A of S, with uniform velocities $u\ kmh^{-1}$ and $\sqrt{3}u\ kmh^{-1}$ in directions east of north making angles 60° and 30° respectively, at the same instant. Draw in a single diagram, the velocity triangles for the motion of B_1 and B_2 relative to S.
Hence, If the range of the guns on ship S is 6.5 km, for how long boat B_1 is exposed to attack?
Also, show that the shortest distance between S and B_1, S and B_2 occur at the same instant, and show that at this instant the distance between B_1 and B_2 is $2\sqrt{3}$ km.

(12)

(a) The triangle ABC shown in the figure is a vertical cross section through the centre of gravity of a smooth uniform wedge of mass M. There is a smooth rail DE parallel to BC in the wedge. The line AB and AC are the line of greatest slope of the face containing it and $A\hat{B}C = \frac{\pi}{3}$. The wedge is placed with the face containing BC on a smooth fixed horizontal table. A particle P of mass m is kept on DE. A light inextensible string is attached to P at one end, and is passed over small smooth pulleys at A and E, the other end being attached to a fixed point H. Here A and H are on ~~a~~ same horizontal level. A particle Q of mass 2m is placed at a point on AB, and the system is released from rest with the string staut.
Show that the magnitude of acceleration of wedge is equal to the magnitude of acceleration of P relative to wedge. Show that the acceleration of wedge is $\dfrac{\sqrt{3}\,mg}{2M + 11m}$.

(b) As shown in the figure, a semicircular smooth thin tube of radius a has another semicircular smooth pulley of radius a fitted to it, and is held fixed in a vertical plane. A sphere P of mass 2m is placed at point A on the horizontal diameter, and a sphere Q of mass m is placed at the lowest point B in the tube, and they are connected by a string of length $\frac{3\pi a}{2}$ passing through the tube.
The system is released from rest.
Show that the velocities v of the particles are given by $v^2 = \frac{2ga}{3}\left[2\theta + \cos\theta - 1\right]$ when the radius through Q has rotated an angle θ $\left(0 < \theta < \frac{\pi}{2}\right)$ with the downward vertical. Find the reaction between Q and tube at this instant.
Then show that at a certain instant during the motion, the reaction between the tube and Q vanishes, and find the tension in the string at this instant.

(13) A particle of mass 2m is suspended from a fixed point O by a light elastic string of natural length l, it is in equilibrium at C, where the extension is l. Show that the modulus of elasticity of the string is 2mg.
Now, the particle of mass 2m is removed, and a particle P of mass m is attached to the end of the string. Another particle Q of mass m is attached to P by a light inextensible string PQ as shown in the diagram. The particle P is held at rest at C and then a force is applied downwards, pulling it down a distance l, and released. Show that both strings are taut and the extension of the elastic string is $x + l$, then $\ddot{x} + \frac{g}{l}x = 0$.

Using the formula $\dot{x}^2 = \omega^2(c^2 - x^2)$, find the amplitude C of this motion.
The PQ string is cut at the instant when the particle P reaches the point A, which is at a distance l below O.
Show that the equation of motion of the particle P after this is given by $\ddot{y} + \frac{2g}{l}\left(y - \frac{l}{2}\right) = 0$. where $AP = y$.
Assuming the solution of the above equation is of the form $y = \frac{l}{2} + \alpha\cos\omega t + \beta\sin\omega t$, find the values of the constant α, β and ω.
Hence, find the centre and amplitude of the simple harmonic motion executed by the particle as it moves from A to B.
Find the velocity of particle when it is at point D which is at a depth of $\frac{7l}{4}$ from O. Show that the particle reaches D after a time $\frac{\pi}{3}\sqrt{\frac{l}{g}}(3+\sqrt{2})$ from the instant of release.

(14)
(a) Let $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OB} = \underline{b}$ and $\overrightarrow{OC} = 2\underline{a}$. where $\underline{a}$ and $\underline{b}$ are two non-zero and non-parallel vectors. The point D is on AB such that $AD : DB = 2:1$. The extended OD line meets BC at E. Show that $3\overrightarrow{OD} = \underline{a} + 2\underline{b}$. By taking $\overrightarrow{OE} = \lambda\overrightarrow{OD}$ and $\overrightarrow{BE} = \mu\overrightarrow{BC}$ for $\lambda, \mu \in \mathbb{R}$, show that $(\lambda - 6\mu)\underline{a} + (2\lambda + 3\mu - 3)\underline{b} = 0$. Hence, show that $5\overrightarrow{OE} = 6\overrightarrow{OD}$.
If the lines OA and AE are perpendicular to each other, show that the angle between the vectors $\underline{a}$ and $\underline{b}$ is $\cos^{-1}\left(\frac{3|\underline{a}|}{4|\underline{b}|}\right)$.

(b) The co-ordinates of the points A, B and C respect to the rectangular cartesian axes OX and OY are $(2,0)$, $(3,\sqrt{3})$ and $(0, 2\sqrt{3})$ respectively. Forces of magnitudes $6\sqrt{3}$, 4, $4\sqrt{3}$ and λ newton act along AB, BC, CA and CO respectively, in the directions indicated by the order of the letters.
If the resultant of this system of forces makes an angle $\pi/6$ anticlockwise with the positive direction of the x-axis, Show that $\lambda = 2$. Also, find the magnitude of the resultant force and the co-ordinate of the point where its line of action intersects the x axis.
Now, a couple of magnitude 24 Nm acting clockwise is added to this system of forces. Find the equation of line of action of new force system.

15 (a) Four uniform rods AB, BC, CD and DA, each of weight w but of different lengths are smoothly jointed at their ends to form a rectangle ABCD and are held in place by a light rod joining A to C. $AB = CD = 4a$ and $AD = BC = 3a$. The vertex A is smoothly hinged to a fixed point, allowing the rectangle to rotate freely in a vertical plane. The rectangle is held with AC horizontal and B below AC by a force P acting at C along CD in the plane of the rectangle. Show that $P = \frac{10w}{3}$ and ~~the componen~~

Find the component of reactions along the joints AD and CD. Find the thrust in the light rod AC.

(b) Framework shown in the figure consists of five light rods AB, BC, AD, AC and CD freely jointed at their ends. $AD = a$, $A\hat{D}C = 60^\circ$ and $A\hat{C}B = C\hat{A}B = 30^\circ$ and a load of $4w$ is suspended at D and a ... of $2P$ is suspended at B. It is hinged at A to a fixed point and is in equilibrium in a ~~horizontal~~ vertical plane with AD horizontal. Find the value of P in terms of w. Using Bow's notation, draw a stress diagram and hence, find the stresses in the five rods, stating whether they are tensions or thrusts.

(16) Show that the centre of mass of a uniform solid right circular cone of base radius r and height h is at a distance $h/4$ from the centre of base. A solid hemisphere of centre O and radius $2a$ has an object S of centre C and radius a cut off by a plane perpendicular to its axis, as shown in the figure.
Show by using integration, that its centre of mass is at a distance $\frac{3a}{4(16-9\sqrt{3})}$ from O on the axis.
A cone of radius a, height $\sqrt{3}a$ and a uniform density is attached to the circular face of the S as shown in the adjacent figure, so that their axes of symmetry coincide.
Show that the centre of mass of the composite body thus formed is at a distance $\frac{3a}{8}(2+\sqrt{3})$ from the centre of the O along the axis of symmetry. Find the weight to be hung from the vertex B so that the axis is horizontal when the composite body is freely suspended from the point A.

(17)

(a) Let A and B be two events in a sample space. In usual notation, If A' and B' are independent, Show that,

(i) A and B are independent.

(ii) A' and B are independent.

(b) Boxes A, B and C are identical and contain red and white balls as shown in the table below. An unbiased die is thrown and a box is selected according to the number on the uppermost face.

Box	Red ball	White ball	the no. on the face of the die
A	3	5	1, 2, 3
B	4	4	4, 5
C	5	3	6

A die is thrown, a box is selected and a ball is drawn, find the probability that the drawn ball is white.
Given that the selected ball is white, find the probability that it is from box B.

(c) The heights of 100 students in a school were measured to the nearest centimeter and the results are as follows,

height (cm)	150-154	155-159	160-164	165-169	170-174
No. of students	9	17	35	23	16

Estimate the mean μ and standard deviation σ of this distribution. Find the coefficient of skewness defined as $k = \frac{\mu - M_0}{\sigma}$. Here M_0 is the mode.